

Metadata Description for Supplementary Materials to paper ‘Rotational Penetration in Sand: Experiments, Modeling, and Implications’

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ABSTRACT

The file `metadata.csv` provides the test-condition metadata for the rotational penetration experiments.

METADATA DESCRIPTION

Each row corresponds to one unique test condition, identified by `test`. Each test condition was repeated in three replicate penetration trials; the metadata file describes the prescribed testing condition, while the corresponding raw trial files contain the time-series force and torque measurements.

The metadata file contains 195 unique test conditions covering 15 testing groups, including shallow, deep, and top-surge penetration series. The principal kinematic variable is the

tangential-to-axial velocity ratio, denoted as $u = v_r/v_v$ in the manuscript and stored as `rs` in the metadata file.

The columns are defined as follows:

test Unique test-condition identifier used to link metadata to raw trial files.

material Granular material used in the test.

density Density state of the specimen, reported as loose or dense.

dr Average relative density of the specimen, in percent.

penetrator Penetrator geometry or cross-sectional shape.

d Nominal penetrator diameter, in mm.

rpm Prescribed rotational speed, in revolutions per minute.

vr Circumferential velocity at the penetrator perimeter, in m/s.

vv Vertical penetration velocity, in m/s.

vt Resultant velocity of the penetrator, in m/s.

rs Relative slip ratio, equivalent to $u = v_r/v_v$.

depth Target penetration depth normalized by penetrator diameter, such as 8D, 13D, or 25D.

container Container identifier used for the testing setup.

b_c Container diameter, B_c , in mm.

bc_d_ratio Container-to-penetrator diameter ratio, B_c/D .

group_name Descriptive testing-group name encoding material, density, geometry, depth, and setup.

group Numeric testing-group identifier.

subgroup Identifier for paired tests within a group, generally corresponding to tests with comparable resultant velocity and the same reference condition.

The non-rotational reference tests are identified by $\text{rpm} = 0$, $\text{vr} = 0$, and $\text{rs} = 0$. Rotational tests within the same `group_name` and `subgroup` are paired with the corresponding non-rotational reference to calculate force and energy ratios. The `group_name` and `group` columns are used for

49 group-level analysis and plotting, while subgroup is used to preserve appropriate reference-test
50 pairing during data reduction.